

Module 4: A+ Troubleshooting and Helpdesk

DAY 1:-

Task 1:

1. Dead PC that won't power on

A multimeter checks the PSU's output voltages (e.g., testing the 24-pin ATX connector for +3.3V, +5V, +12V) to confirm whether the power supply itself is faulty before replacing any other component.

2. Phone charging port not working

A multimeter tests continuity across the charging port's solder points on the phone's logic board to identify a broken trace or cold solder joint preventing power from reaching the battery.

3. Laptop battery not charging

A multimeter measures the battery's actual voltage output directly at its terminals to determine whether the battery itself has degraded/died, versus the issue being with the charging circuit or adapter.

Task 2:

POST Card motherboard test

- POST Card is inserted in the PCI slot on the motherboard.
- POST Card displays error code. Through error code you can diagnose the problem and find the solution.

Some of the Error Codes, faults and Solutions it shows are:

1. **00Ac** - All Okay.
2. **d454** - PC shows no display : **Fault** - RAM : **Solution** - Change or clean RAM
3. **0000** - PC shows no display : **Fault** - CPU not detected : **Solution** - Check Core (CPU) section.
4. **E098** - PC shows no display : **Fault** - RAM not detected : **Solution** - Check RAM Slots.

Task 3:

HWMonitor ARM64				
File View Help				
Sensor	Value	Min	Max	
AASHANAP				
VBSA				
Utilizations				
Physical Memory Load	78.0 %	77.0 %	81.0 %	
Available Physical Mem...	0.9 GB	0.7 GB	0.9 GB	
Apple silicon				
Utilizations				
Package	99.6 %	0.0 %	100.0 %	
Apple 0x0	99.6 %	0.0 %	100.0 %	
Clocks				
Apple 0x0 (Max)	2001 MHz	1999 MHz	2089 MHz	
Core #0	2001 MHz	1996 MHz	2089 MHz	
Core #1	2001 MHz	1998 MHz	2046 MHz	
VMware Virtual NVMe Disk				
Utilizations				
Used Space (c:)	53.4 %	53.4 %	53.4 %	
Available Capacity (c:)	29.4 GB	29.4 GB	29.4 GB	
Activity	7.8 %	0.0 %	75.6 %	
Temperatures				
Assembly	11759.0 °C	11759.0 °C	11759.0 °C	
Transfers				
Read Rate	6.64 MB/s	0.00 MB/s	14.93 MB/s	
Write Rate	10.37 MB/s	0.00 MB/s	41.52 MB/s	
Counters				
Health Status	100 %	100 %	100 %	

Task 4:

Tool	What it Checks	Example Problem Detected
Multimeter	Physical voltage output at power connectors, continuity across circuits/ traces.	PSU's 24-pin ATX connector reads 0V on the +12V rail, confirming a dead power supply as the cause of a PC not turning on.
POST Card	Hexadecimal status codes during the boot sequence, before the OS or even a display loads.	A code corresponding to "memory initialization failure" appears and the system hangs, pointing to a faulty or improperly seated RAM module.
Software Diagnostic Tools	Real-time sensor data and drive health once the OS has booted — temperatures, voltages, fan speeds, S.M.A.R.T. values.	CrystalDiskInfo shows a "Caution" health status on an SSD due to a rising reallocated sector count, indicating the drive is beginning to fail.

